



CANCER EPIGENETICS COURSE

PACKAGES AND BASIC VISUALIZATION IN R

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Packages

- R is a functional system that can be extended through packages, which store groups of related software.
- CRAN: The Comprehensive R Archive Network is the central repository where users contribute thousands of packages
- Ex: ggplot2, dplyr, tidyr, tidyverse,...



BiocManager

- BiocManager: used to manage and install packages from Bioconductor
- Standard R packages are usually hosted on CRAN (installed via `install.packages()`)
- Bioconductor packages require their own specialized manager to ensure version synchronization between R and the Bioconductor repository.
- How to use:
 - `install.packages("BiocManager")`
 - `BiocManager::install("DESeq2")`

Bioconductor/ BiocManager



CRAN Package For Managing Bioconductor Packages

16

Contributors

20

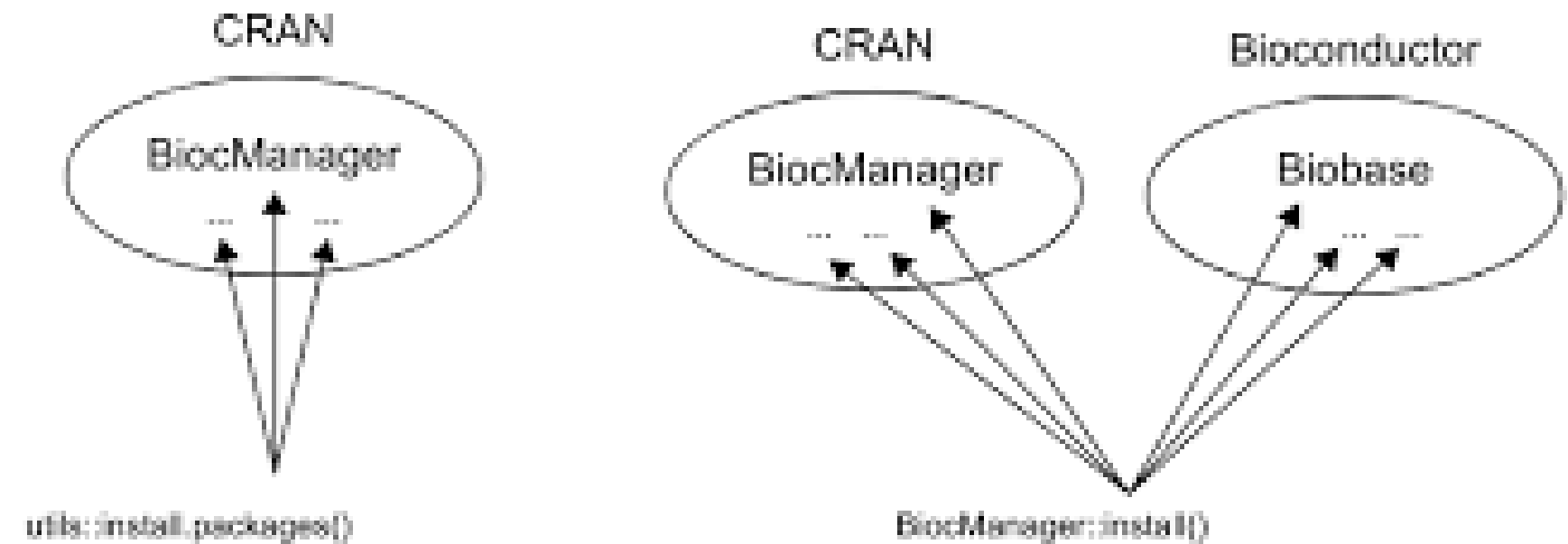
Issues

75

Stars

27

Forks

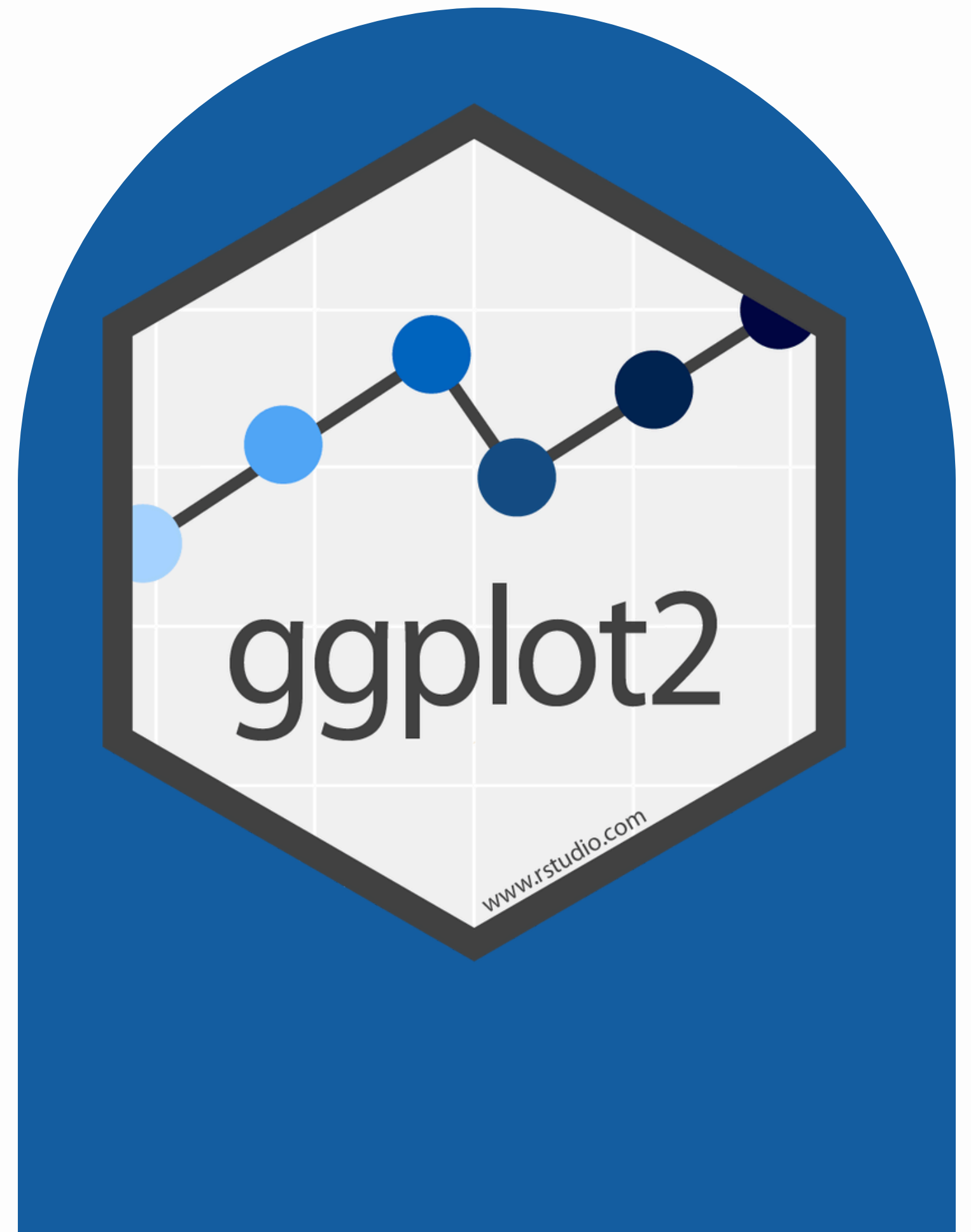


ggplot2

- Plotting package in R
- `install.packages("ggplot2")`
- `library(ggplot2)`
- Syntax:

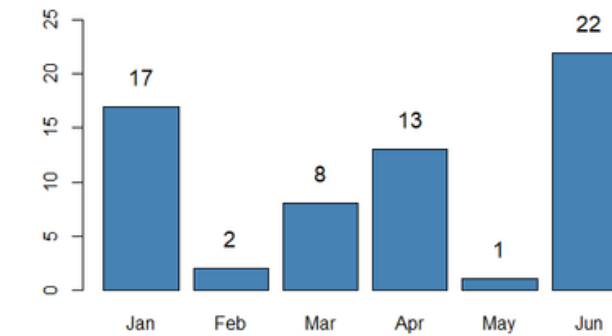
`ggplot(<data>, x=<x_variable>, y=
<y_variable>) + geom_<action> + ...`

#geom_action consist of many functions

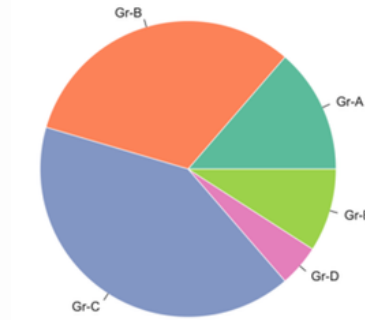


Data visualization

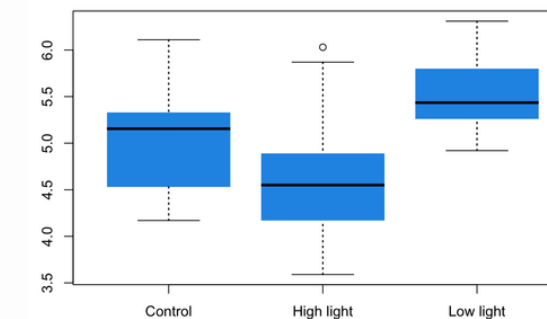
- Visualize data as graphs
- Type of graphs:
 1. Univariate (single variable): bar chart, pie chart, histogram, density,...
 2. Multivariate (many variables): scatter plots, heatmaps,...



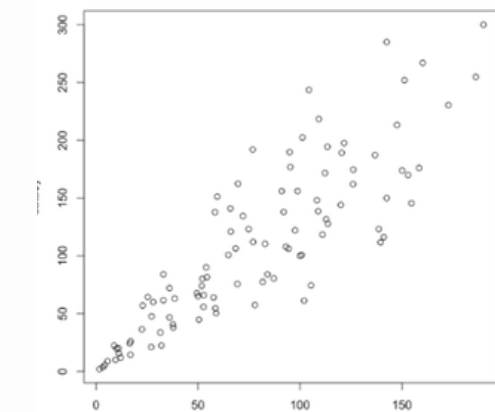
Bar chart



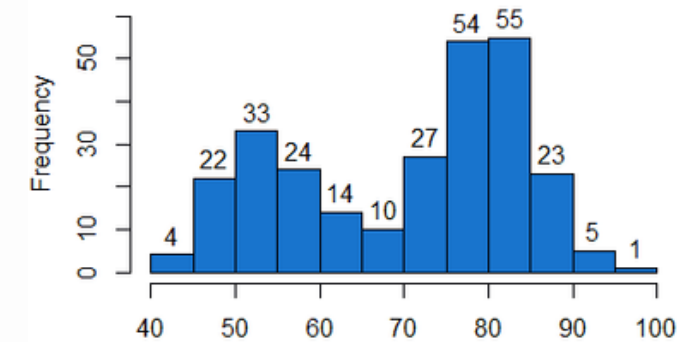
Pie chart



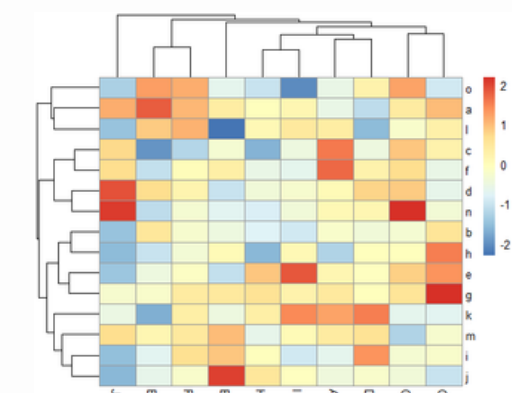
Box plot



Scatter plot



Histogram



Heatmap

When to use each graph

Graph	Variable
Histogram	Numerical univariate
Density	Numerical univariate
Box plot	Numerical univariate
Bar chart	Categorical univariate
Pie chart	Categorical univariate
Scatter plot	Numerical multivariate
Heatmap	Numerical multivariate

When to use each graph

Ex1: Counting how many students in a class prefer different fruits (e.g., 10 like Apples, 5 like Bananas, 8 like Oranges).

Ex2: Showing how you spend your monthly budget (e.g., 50% Rent, 30% Food, 20% Savings).

Ex3: Mapping the heights of 100 people to see if most are "average" height or if there are more tall/short people.

Ex4: Looking at the time of day people visit a website to see the "peak" traffic hours as a smooth wave.

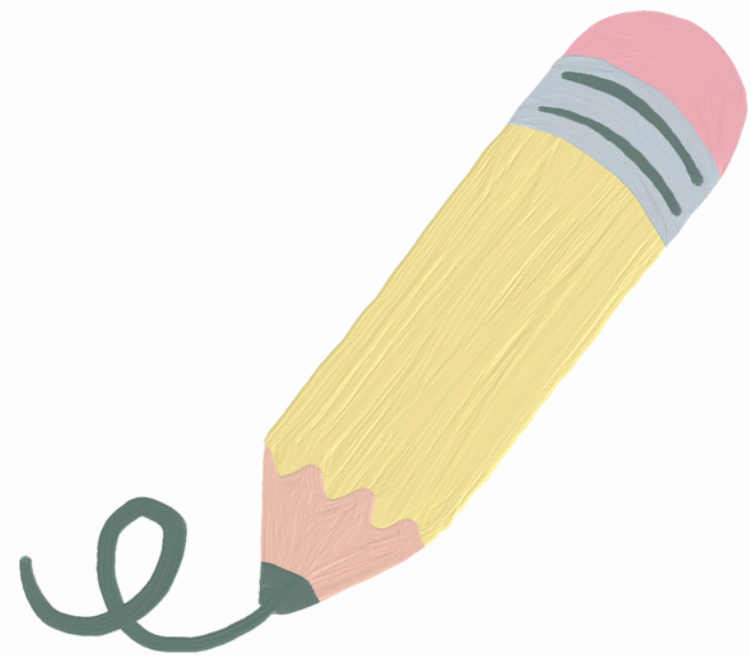
Ex5: Comparing test scores between two different schools to see which school has a wider spread of grades.

Ex6: Plotting Temperature vs. Ice Cream Sales. As the temperature goes up, do sales also go up?

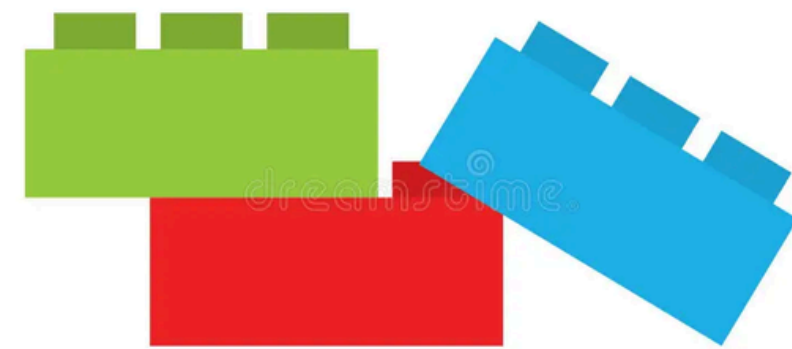
Cheat sheet

If you want to...	Use a...
Compare simple categories	Bar Chart
Show percentages of a total	Pie Chart
See the "spread" of scores	Histogram or Box Plot
Check if X affects Y	Scatter Plot
Find patterns in a big grid	Heatmap

How to visualize data in R?



**R-base
approach**



**ggplot2
approach**